Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:

In this task, the **IGN dataset (ign.csv)** was processed and prepared for analysis using the **Pandas library in Python**. The objective was to clean and structure the dataset so that it becomes suitable for further data analysis and visualization.Initially, the dataset was loaded into a Pandas DataFrame, and its structure was explored by displaying the first and last few rows along with its overall shape. This helped in understanding the number of records and features present in the dataset.

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

```
   Unnamed: 0 score_phrase  ... release_month release_day
0           0      Amazing  ...             9          12
1           1      Amazing  ...             9          12
2           2        Great  ...             9          12
3           3        Great  ...             9          11
4           4        Great  ...             9          11

[5 rows x 11 columns]
       Unnamed: 0 score_phrase  ... release_month release_day
18620       18620         Good  ...             6          29
18621       18621      Amazing  ...             6          29
18622       18622     Mediocre  ...             6          28
18623       18623  Masterpiece  ...             6          28
18624       18624  Masterpiece  ...             6          28

[5 rows x 11 columns]
(18625, 11)
score        0
genre       36
platform     0
dtype: int64
```

Figure 1.1 : Displaying outputs

Conclusion:

In this task, the IGN dataset was successfully cleaned and prepared for analysis using **Pandas**. The dataset was loaded and explored to understand its structure, and unnecessary data such as the **"Unnamed: 0"** column was removed to improve clarity. Missing values in important columns were handled appropriately—numerical values in the **score** column were replaced with the mean, while categorical values in the **genre** (and optionally platform) column were filled using the mode.

Scenario 2: Line Graph (Score Trend) + Save (□ Easy)

Description:

In this task, the IGN dataset was analyzed to understand how **average game scores have changed over time**. The process began by grouping the dataset based on the **release_year** column, which allowed aggregation of data year-wise. Using this grouping, the **average score for each year** was calculated with the help of Pandas.Next, the resulting data (years and their corresponding average scores) was converted into **NumPy arrays**. This step ensures compatibility with numerical operations and makes it easier to plot the data efficiently.

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph :

The graph indicates that while average game scores dropped in earlier decades, they have **consistently improved in recent years**, suggesting better game quality and development standards over time. The X-axis represents the release year, while the Y-axis represents the average score.

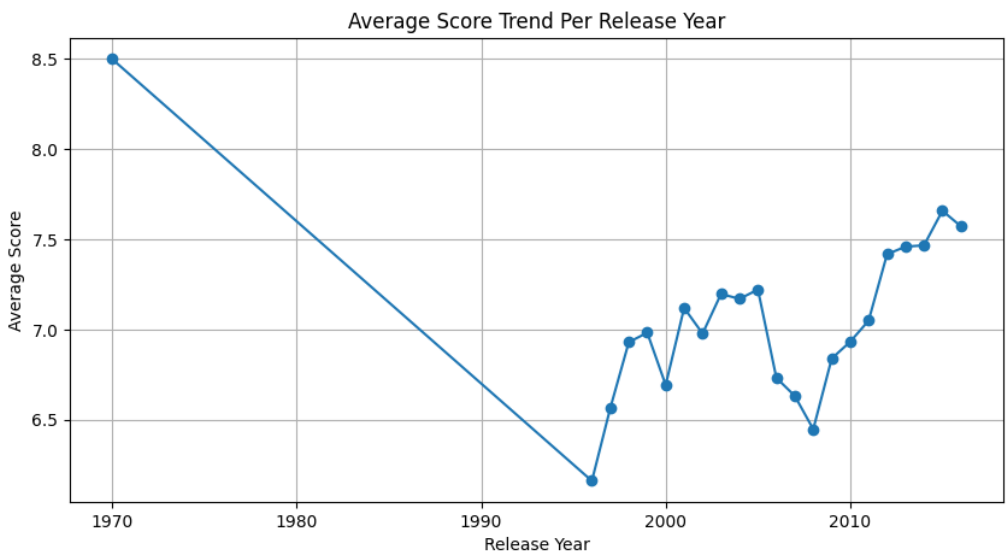

Figure 2.1: Line Chart of Every Year's Average Score

Conclusion:

The dataset was grouped by release year to calculate the average game score for each year. These values were converted into NumPy arrays and visualized using a line graph. The resulting graph illustrates how average game scores have changed over time. Based on the trend, it can be observed whether game quality (as reflected by scores) has improved, declined, or remained stable across different years. This analysis provides valuable insight into the evolution of game ratings over time.

Scenario 3: Filtering + Bar Chart + Save (□ Medium)

Description:

In this task, the IGN dataset was analyzed to identify which gaming platforms have the highest number of **high-rated games**. The process began by filtering the dataset to include only those games with a **score greater than 7**, focusing on well-reviewed titles.Next, the filtered data was grouped by the **platform** column, and the number of high-rated games for each platform was counted using Pandas. From this result, the **top 10 platforms** with the highest counts were selected to simplify the analysis and focus on the most significant contributors.

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph :

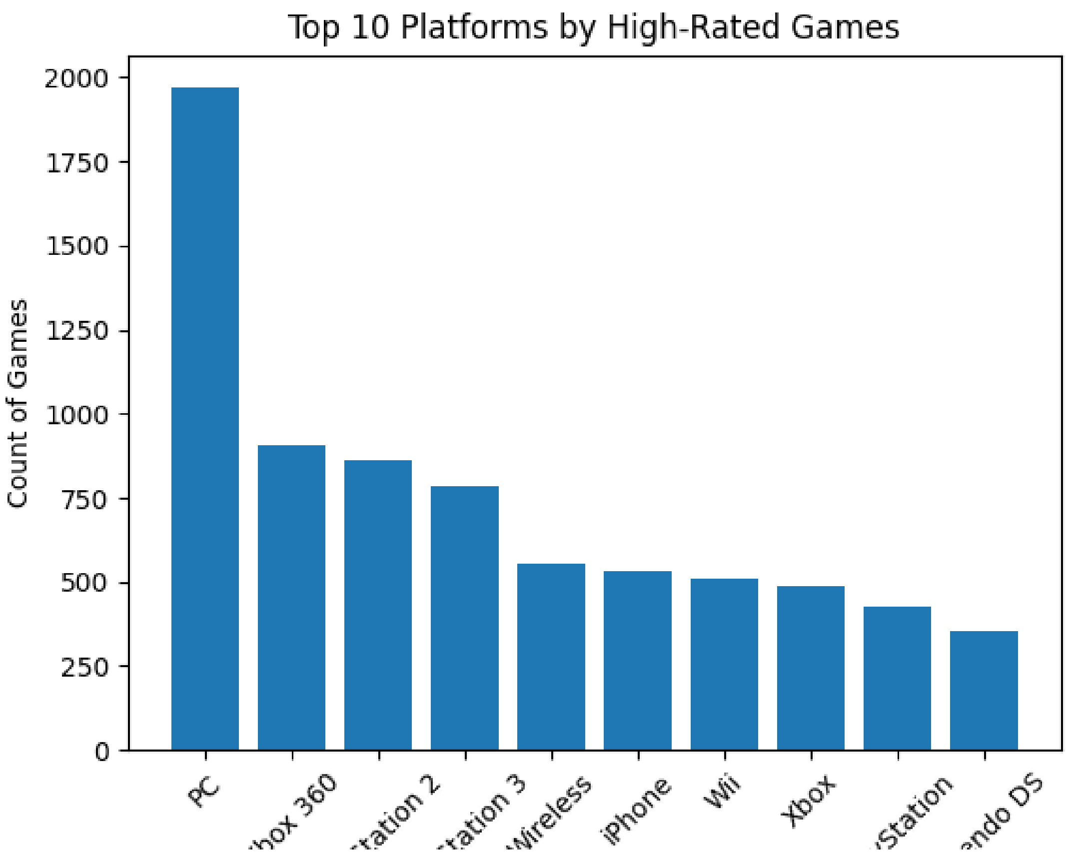

Figure 3.1: Bar Chart of Top 10 Platforms

Conclusion:

The analysis filtered games with scores greater than 7 and counted their distribution across different platforms. The bar chart highlights the top 10 platforms with the highest number of well-rated games. This shows which platforms are most associated with high-quality game releases. Platforms appearing at the top of the chart can be considered more successful in delivering highly rated games.

Scenario 4: Aggregation + Pie Chart + Save (□ Medium)

Description:

In this task, the IGN dataset was analyzed to understand the **distribution of games across different genres**. From this distribution, the **top 5 most common genres** were selected to focus the analysis on the most significant categories. This helps simplify the visualization and highlights the dominant genres in the dataset.A **pie chart** was plotted using Matplotlib, where each slice represents a genre and its proportion relative to the total. Percentage labels were added using the **autopct parameter** (e.g., '%.1f%%') to clearly show each genre's contribution.

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph :

The pie chart reveals that **Action games are by far the most popular genre**, while other genres like Sports and Shooter also contribute significantly. However, Racing and Adventure games have a comparatively smaller presence in the dataset.

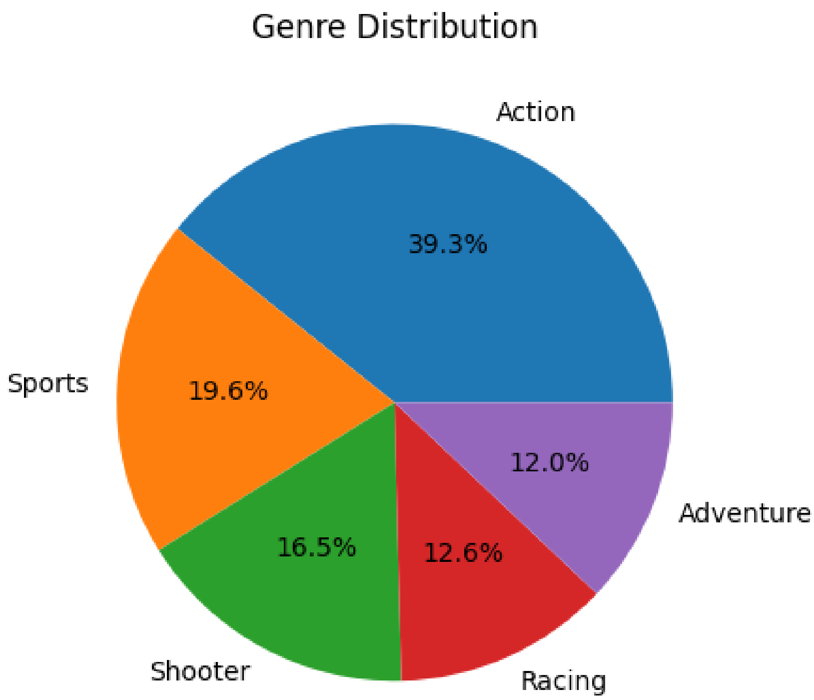

Figure 4.1: Pie Chart of Genre Distribution

Conclusion:

The dataset was analyzed to count the number of games in each genre, and the top 5 genres were selected for visualization. A pie chart was created to display the percentage contribution of each genre. The chart clearly highlights the most dominant genres in the dataset, showing which types of games are most frequently produced and reviewed.

Scenario 5: Advanced Analysis + Multiple Graphs (⚫ Hard)

Description:

In this task, the IGN dataset was analyzed to explore **game score patterns, trends over time, and quality distribution** using **Pandas, NumPy, and Matplotlib**. a new column called **score_category** was created to classify games into three groups based on their scores: *Excellent* (score ≥ 9), *Good* ($7 \leq$ score < 9), and *Average* (score < 7). This categorization helps simplify the analysis by grouping games according to their quality levels. Additionally, the **editors_choice** column was converted from categorical values ('Y' and 'N') into numerical values (1 and 0) to enable easier analysis and comparison. Overall, this task demonstrates how to **engineer features, analyze trends, and visualize patterns** to gain meaningful insights from a real-world dataset.

Task	Description
1.	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph:

Line Graph (Score Growth):

Shows how average game scores change year by year. It highlights fluctuations and trends in review scores over time.

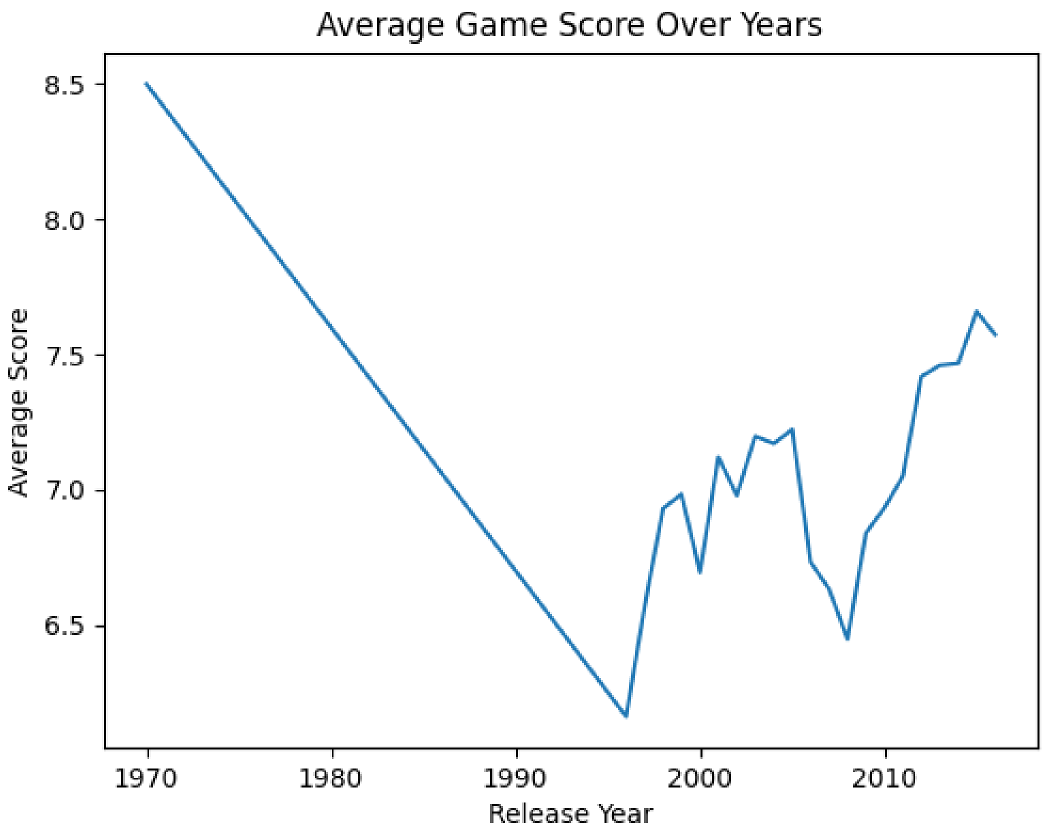

Figure 5.1: Line Graph of Yearly Score Growth

Stack Bar Chart:

Displays the count of "Excellent", "Good", and "Average" games per year. It helps compare how score categories evolve over time.

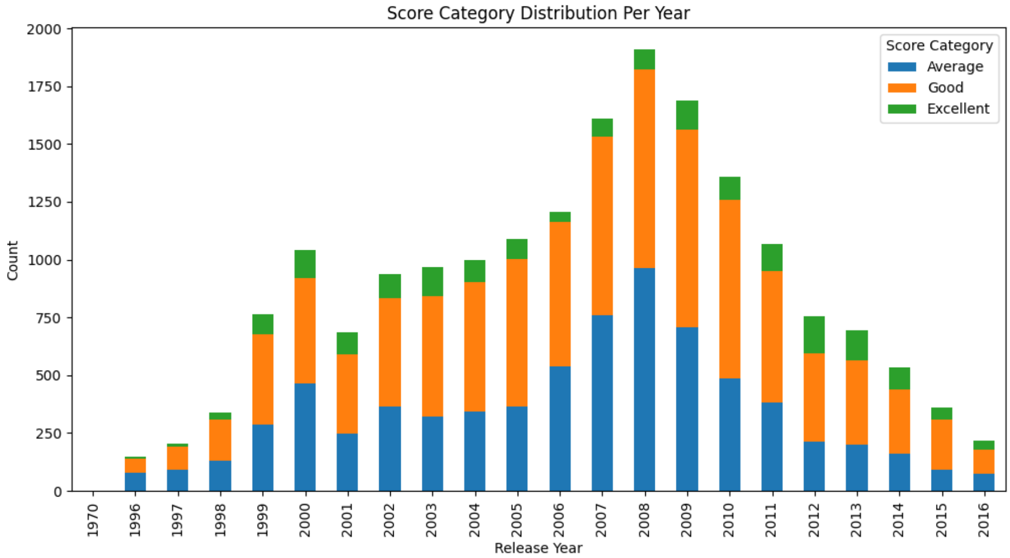

Figure 5.2: Stacked Bar Chart of Score Category per Year

Histogram:

Shows the overall distribution of game scores, indicating how frequently different score ranges occur.

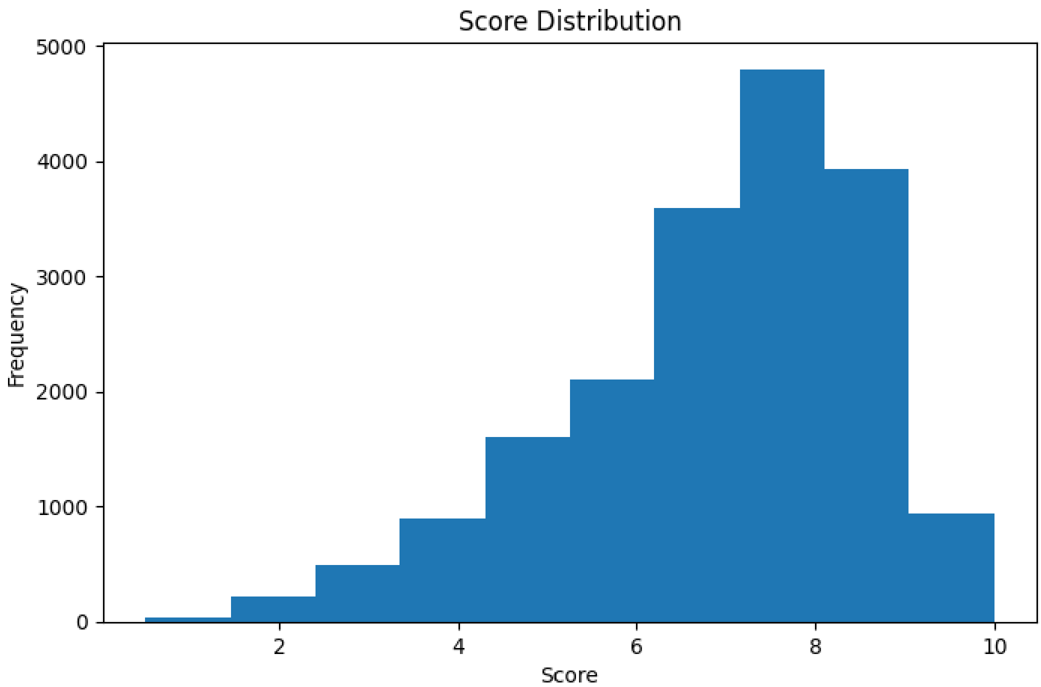

Figure 5.3: Histogram of Score Distribution

Conclusion:
The histogram shows that game scores are concentrated in the higher range, with most games receiving scores between 6 and 9. This indicates that the majority of games are generally well-rated, while very low and very high scores occur less frequently. Additionally, the analysis shows a strong relationship between high scores and editor's choice selections.

Final Conclusion:

This project analyzed the IGN dataset using **Pandas, NumPy, and Matplotlib** to explore patterns in game scores, genres, platforms, and trends over time. Through systematic data cleaning, transformation, and visualization, meaningful insights about the gaming industry were uncovered.The analysis revealed that **average game scores have generally improved and stabilized over time**, indicating advancements in game development and quality standards. Most games consistently fall within the **"Good" score range (7–9)**, while **"Excellent" games are relatively rare**, showing that achieving top-tier ratings is difficult.

What you have learned

This project gave me hands-on experience in **data analysis using Python**, especially with **Pandas, NumPy, and Matplotlib**, and helped me understand how to turn raw data into meaningful insights. I learned how to use Python tools to analyze data effectively, identify trends, and communicate insights through visualizations—skills that are essential for data science and real-world problem solving.

- Learned how to **load and explore datasets** (`head()`, `tail()`, `shape`)
- Removed unnecessary columns and handled **missing values**
- Converted columns into correct **data types**
- Created new features like **score_category**
- Filtered and grouped data for analysis
- Work with real-world datasets
- Think analytically and interpret results
- Present findings clearly for reports or assignments